

Exploring Distributions

STA 310: Homework 2

Instructions

- Write all narrative using full sentences. Write all interpretations and conclusions in the context of the data.
- Be sure all analysis code is displayed in the rendered pdf.
- If you are fitting a model, display the model output in a neatly formatted table. (The `tidy` and `kable` functions can help!)
- If you are creating a plot, use clear and informative labels and titles.
- Render and back up your work regularly, such as using Github.
- When you're done, we should be able to render the final version of the Rmd document to fully reproduce your pdf.
- Upload your pdf to Gradescope. Upload your Rmd, pdf (and any data) to Canvas.

These exercises come from BMLR or are adapted from BMLR, Chapter 3.

Exercises

Exercise 1

At what value of p is the variance of a **binary random variable** smallest? When is the variance the largest? Back up your answer empirically or mathematically.

Mathematical Proof

Let $X \sim \text{Bernoulli}(p)$, where $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

The variance of a Bernoulli random variable is

$$\text{Var}(X) = p(1 - p).$$

Rewrite this expression as a quadratic in p :

$$\text{Var}(X) = p - p^2 = -(p^2 - p).$$

Complete the square:

$$\text{Var}(X) = -\left[\left(p - \frac{1}{2}\right)^2 - \frac{1}{4}\right] = -\left(p - \frac{1}{2}\right)^2 + \frac{1}{4}.$$

Since $\left(p - \frac{1}{2}\right)^2 \geq 0$ for all $p \in [0, 1]$, it follows that

$$\text{Var}(X) \leq \frac{1}{4}.$$

Equality holds when

$$p = \frac{1}{2},$$

so the variance is maximized at $p = \frac{1}{2}$ with

$$\text{Var}(X) = \frac{1}{4}.$$

To find the minimum, note that

$$p(1 - p) = 0 \iff p = 0 \text{ or } p = 1.$$

Thus, the variance is minimized at $p = 0$ or $p = 1$, where

$$\text{Var}(X) = 0.$$

Empirical Check in R

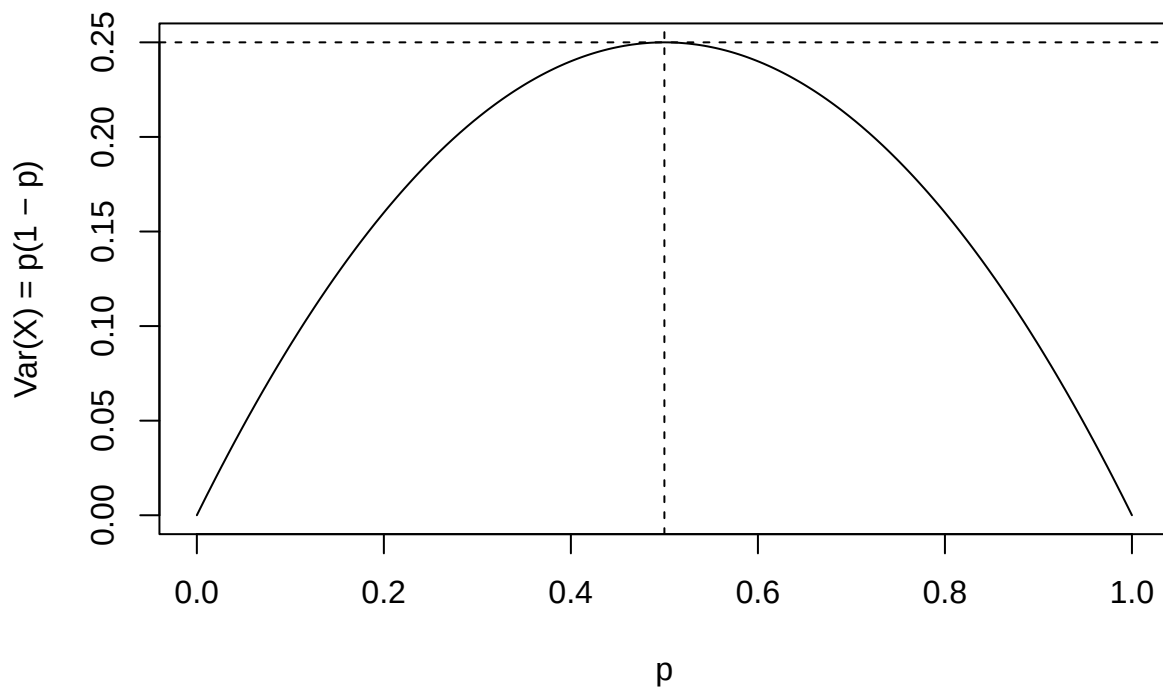
```
p <- seq(0, 1, by = 0.001)
# Create a sequence of probability values from 0 to 1 in increments of 0.001

v <- p * (1 - p)
# Compute the variance of a Bernoulli(p) random variable for each value of p

plot(p, v, type = "l", xlab = "p", ylab = "Var(X) = p(1 - p)")
# Plot variance as a function of p, using a line plot

abline(v = 0.5, lty = 2)
# Add a vertical dashed line at p = 0.5, where the variance is maximized

abline(h = 0.25, lty = 2)
```



Add a horizontal dashed line at variance = 0.25, the maximum possible variance

Exercise 2

How are hypergeometric and binomial random variables different? How are they similar?

Both the binomial and hypergeometric random variables model the number of successes in a fixed number of trials. In each case, every trial results in one of two possible outcomes: success or failure. Consequently, the support of both random variables is the set $\{0, 1, \dots, n\}$.

For both distributions, the expected number of successes is the same. Let p denote the proportion of successes in the population, for both distributions:

$$\mathbb{E}[X] = np.$$

Thus, on average, the two models predict the same number of successes when the underlying success probability is p .

When the population size is large relative to the sample size, the binomial distribution provides a good approximation to the hypergeometric distribution.

The primary difference of the two distributions is in the sampling mechanism. A binomial random variable assumes that sampling is performed *with replacement*. As a result, the trials are independent and the probability of success remains constant across trials. In contrast, a hypergeometric random variable assumes sampling *without replacement*, which introduces dependence between trials and causes the probability of success to change after each draw.

A binomial random variable is defined as

$$X \sim \text{Binomial}(n, p),$$

with probability mass function

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

A hypergeometric random variable is defined as

$$X \sim \text{Hypergeometric}(N, K, n),$$

where N denotes the population size, K denotes the number of successes in the population, and n denotes the sample size. Its probability mass function is

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

Another important distinction between the two distributions is their variances. For a binomial random variable,

$$\text{Var}(X) = np(1 - p).$$

For a hypergeometric random variable, the variance is given by

$$\text{Var}(X) = np(1 - p) \frac{N - n}{N - 1}.$$

The factor $\frac{N-n}{N-1}$, known as the finite population correction, reflects the dependence between trials and causes the hypergeometric variance to be smaller than the binomial variance.

When the population size N is much larger than the sample size n , the finite population correction factor satisfies

$$\frac{N - n}{N - 1} \approx 1.$$

In this setting, the hypergeometric distribution is well approximated by a binomial distribution with

$$p = \frac{K}{N}.$$

Exercise 3

How are exponential and Poisson random variables related?

Exercise 4

How are geometric and exponential random variables similar? How are they different?

Exercise 5

A university's college of sciences is electing a new board of 5 members. There are 35 applicants, 10 of which come from the math department. What distribution could be helpful to model the probability of electing X board members from the math department?

Exercise 6

Chapter 1 asked you to consider a scenario where *"The Minnesota Pollution Control Agency is interested in using traffic volume data to generate predictions of particulate distributions as measured in counts per cubic feet."* What distribution might be useful to model this count per cubic foot? Why?

Exercise 7

Chapter 1 also asked you to consider a scenario where *"Researchers are attempting to see if socioeconomic status and parental stability are predictive of low birthweight. They classify a low birthweight as below 2500 g, hence our response is binary: 1 for low birthweight, and 0 when the birthweight is not low."* What distribution might be useful to model if a newborn has low birthweight?

Exercise 8

Chapter 1 also asked you to consider a scenario where *"Researchers are interested in how elephant age affects mating patterns among males. In particular, do older elephants have greater mating success, and is there an optimal age for mating among males? Data collected includes, for each elephant, age and number of matings in a given year."* Which distribution would be useful to model the number of matings in a given year for these elephants? Why?

Exercise 9

Describe a scenario which could be modeled using a gamma distribution.

Exercise 10

Beta-binomial distribution. We can generate more distributions by mixing two random variables. Beta-binomial random variables are binomial random variables with fixed n whose parameter p follows a beta distribution with fixed parameters α, β . In more detail, we would first draw p_1 from our beta distribution, and then generate our first observation y_1 , a random number of successes from a binomial (n, p_1) distribution. Then, we would generate a new p_2 from our beta distribution, and use a binomial distribution with parameters n, p_2 to generate our second observation y_2 . We would continue this process until desired.

Note that all of the observations y_i will be integer values from $0, 1, \dots, n$. With this in mind, use `rbinom()` to simulate 1,000 observations from a plain old vanilla binomial random variable with $n = 10$ and $p = 0.8$. Plot a histogram of these binomial observations. Then, do the following to generate a beta-binomial distribution:

- Draw p_i from the beta distribution with $\alpha = 4$ and $\beta = 1$.
- Generate an observation y_i from a binomial distribution with $n = 10$ and $p = p_i$.
- Repeat (a) and (b) 1,000 times ($i = 1, \dots, 1000$).
- Plot a histogram of these beta-binomial observations.

Compare the histograms of the “plain old” binomial and beta-binomial distributions. How do their shapes, standard deviations, means, possible values, etc. compare?

Grading

Total	33
Ex 1	5
Ex 2	5
Ex 3	5
Ex 4	5
Ex 5	2
Ex 6	2
Ex 7	2
Ex 8	2
Ex 9	2
Ex 10	5
Workflow & formatting	3

The “Workflow & formatting” grade is to based on the organization of the assignment write up along with the reproducible workflow. This includes having an organized write up with neat and readable headers, code, and narrative, including properly rendered mathematical notation. It also includes having a reproducible Rmd or Quarto document that can be rendered to reproduce the submitted PDF.